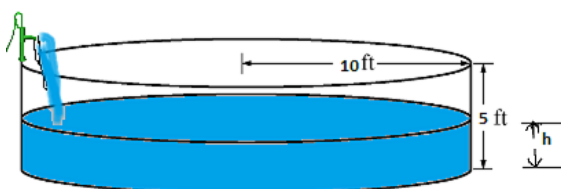


Problem 4

Problem 4

Problem 1 A cylindrical swimming pool with a radius of 10 feet and height of 5 feet is being filled with water. How fast is the water being pumped into the pool (in ft^3/min) if the water level is increasing at a rate of 6 in/minute? Make sure to draw a picture to represent the situation.



Explanation.

$$\frac{dh}{dt} = 6 \text{ in/min} = .5 \text{ feet/min}$$

$$V = \pi r^2 h = \pi(100)h$$

$$\frac{dV}{dt} = 100\pi \frac{dh}{dt}$$

$$\frac{dV}{dt} = 100\pi(.5) = 50\pi \text{ ft}^3/\text{min}$$